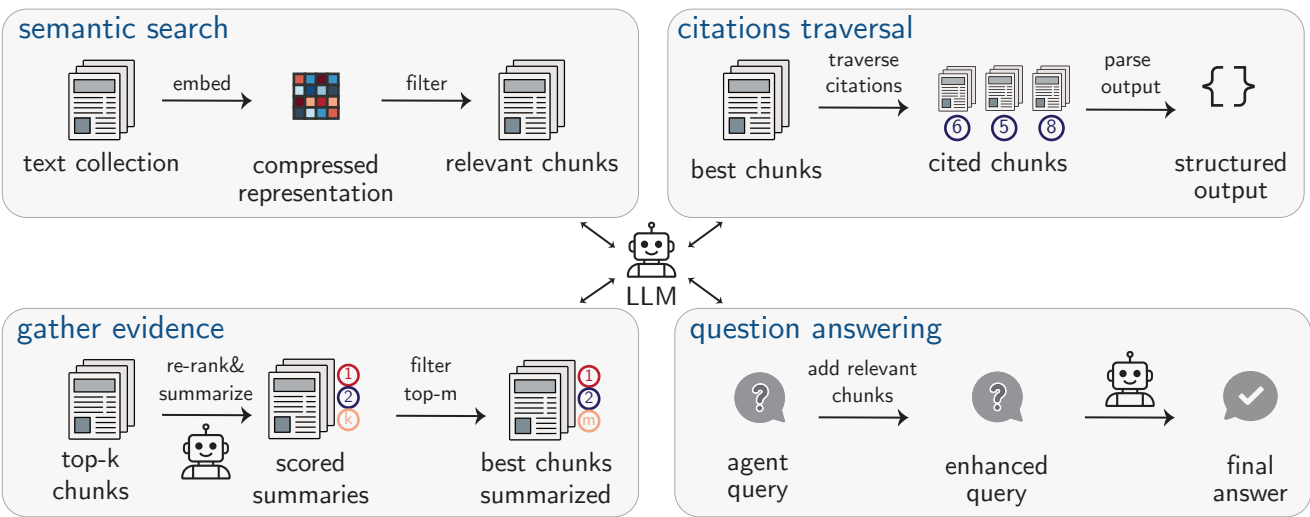


A. agent for knowledge gathering



B. example applications

general questions

Has anyone researched generating new molecules using the mamba architecture?

What is the state-of-the-art DFT functional for organometallic compounds?

structured data extraction

Extract solar cell efficiency information from literature given this schema:

```
class SolarCell:
    material: ...
    architecture: ...
    efficiency: ...
    units: ...
```

C. worked example

query

What is the state-of-the-art (SOTA) of SMILES-to-IUPAC translation using deep learning models?

search top-k (53 papers)

1. Mswahili et al. 2024
2. Zhang et al. 2025
3. Yu et al. 2024
4. Sun et al. 2024
5. Rajan et al. 2024
- ...
53. Seidl et al. 2024

gather evidence (down to 7 papers)

1. Rajan et al. 2024
- A. Used in 1.1.
- Shtopko 2024
- Yu et al. 2024
- Xia et al. 2025
- ...
7. Caldas Ramos et al. 2025

query result

State-of-the-art methods for converting SMILES to IUPAC names now broadly fall into two categories: traditional rule-based deterministic systems and advanced neural network approaches built on transformer architectures (1.1). ...